



Education

M.S. in Computer Science – Data Science & Informatics GPA: 3.7/4.0 Clemson University, Clemson, SC	January 2024 – December 2025
B.E. in Computer Science and Engineering D. Y. Patil College of Engineering and Technology, Kolhapur, India	August 2017 – July 2021

Experience

Researcher – LLM Agents & Human Behavior Clemson University, School of Computing	April 2026 – Present Clemson, SC / Remote
- Researching whether societies of LLM agents can substitute for human participants in behavioral pilot studies. - Reproducing published behavioral experiments and designing new ones with multi-agent frameworks (AgentSociety, YuLan-OneSim, Generative Agents). - Developing a taxonomy and metrics for the behavioral realism, consistency, and human alignment of LLM agents.	
Graduate Teaching Assistant – Applied Data Science Clemson University, School of Computing	Aug 2024 – Dec 2025 Clemson, SC
- Designed Jupyter labs and assignments on data preprocessing, supervised learning (classification, regression), unsupervised learning, model evaluation, and data visualization for a graduate data science course. - Built automated grading pipelines with Python and nbgrader, reducing manual effort and improving scoring consistency. - Maintained live course infrastructure and explained ML concepts and debugged student code during office hours.	
Software Test Engineer Amdocs	Oct 2021 – Dec 2022 Pune, India
- Performed end-to-end, regression, and integration testing on enterprise telecom systems supporting AT&T. - Designed test cases and validation strategies for new features from product requirements and stakeholder input. - Ran functional/UI/API release tests with Selenium and Postman; triaged defects with US/India Agile teams.	

Projects

Movie Recommendation Engine Python, PyTorch, scikit-learn, Flask, React, Docker, AWS EC2	GitHub Live
- Built a data pipeline joining 26M MovieLens ratings with TMDB and IMDb metadata, then compared 7 recommenders (CF, SVD, NMF, NeuMF, hybrid) using a temporal split. - Scaled training to the full dataset by storing the 0.37%-dense rating matrix as 45 MB of sparse CSR (vs. ~42 GB dense) and training a 26.3M-parameter PyTorch NeuMF on 20.7M ratings on a GPU. - Tuned FunkSVD to 0.760 RMSE (21% below item-based CF) and found the lowest-error model isn't the best ranker: TruncatedSVD led on NDCG@10 (0.112 vs. 0.085). - Deployed a cold-start-aware hybrid as a Flask/React app on AWS EC2 with Docker and GitHub Actions CI/CD.	
BB-8: Transformer Language Model Python, PyTorch, Hugging Face, LoRA	GitHub Hugging Face Live
- Built a GPT-style Transformer and BPE tokenizer from scratch in PyTorch and ran 11 tracked experiments, from 112K–5.6M-parameter models to LoRA fine-tuning of Qwen2.5-0.5B on Dolly 15K. - Diagnosed chat failures with an 80-case, 5-configuration test suite, tracing errors to a prompt-format mismatch and a preprocessing bug that truncated 29% of training examples. - Designed an 88-case grounded-QA benchmark whose 80% promotion gate blocked a model with near-perfect perplexity (1.03) that passed only 22% of content and citation checks. - Caught a tokenizer data leak and added dataset hashing, commit pinning, versioned configs, and a model registry.	
BB8 Co-Pilot x Tech Portfolio RAG, Amazon S3 Vectors, OpenAI embeddings, OpenAI API, DynamoDB, AWS Amplify, Next.js, TypeScript	GitHub Live
- Developed a RAG assistant with section-aware chunking, S3 Vectors search, and hybrid semantic-keyword retrieval. - Built a 27-question retrieval benchmark (Hit@3, P95 latency) that gates every merge in CI.	

Skills

Programming	Python, SQL	Data & Visualization	pandas, NumPy, seaborn, matplotlib
ML & Statistics	PyTorch, scikit-learn, Hugging Face Transformers	Tools & Infrastructure	AWS (EC2, S3, Lambda, Amplify, DynamoDB), Docker, Git/GitHub, GitHub Actions (CI/CD)
Core Areas	Machine Learning, Deep Learning, LLMs, GenAI, RAG, MLOps, Computer Vision		